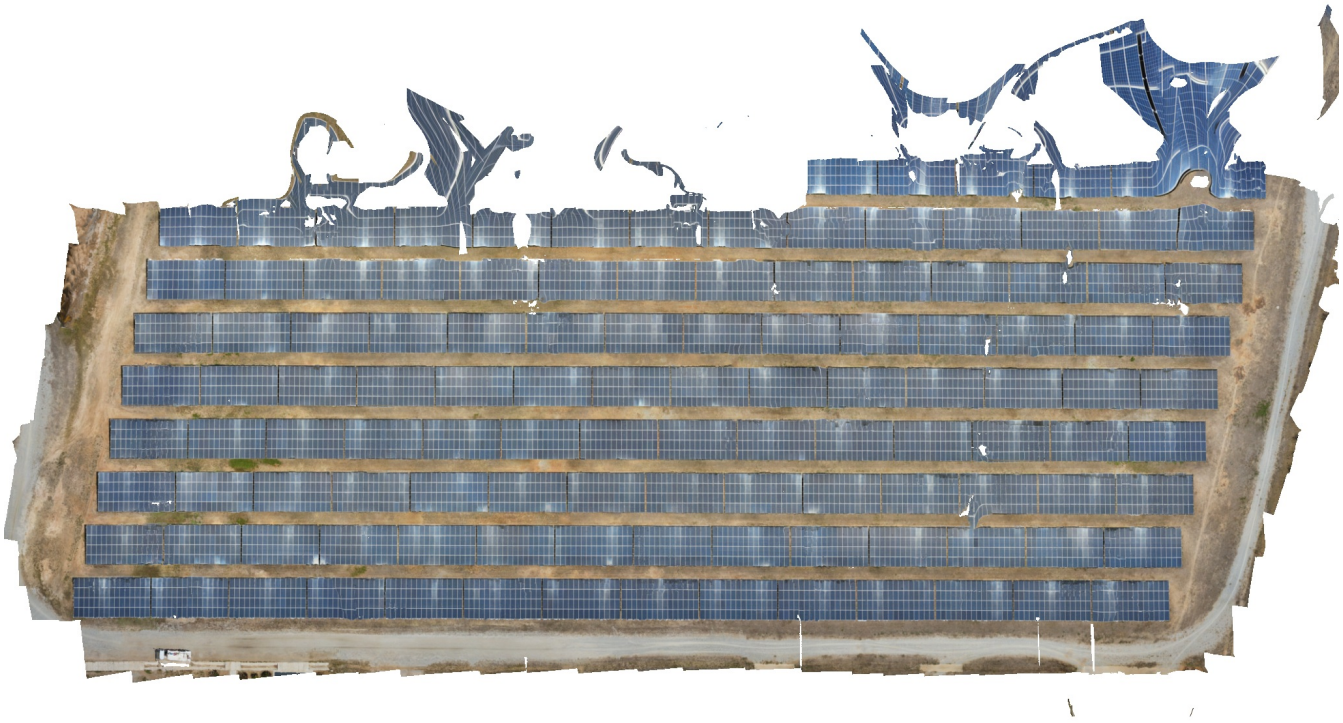
Agisoft Metashape

Processing Report

03 September 2026



Survey Data

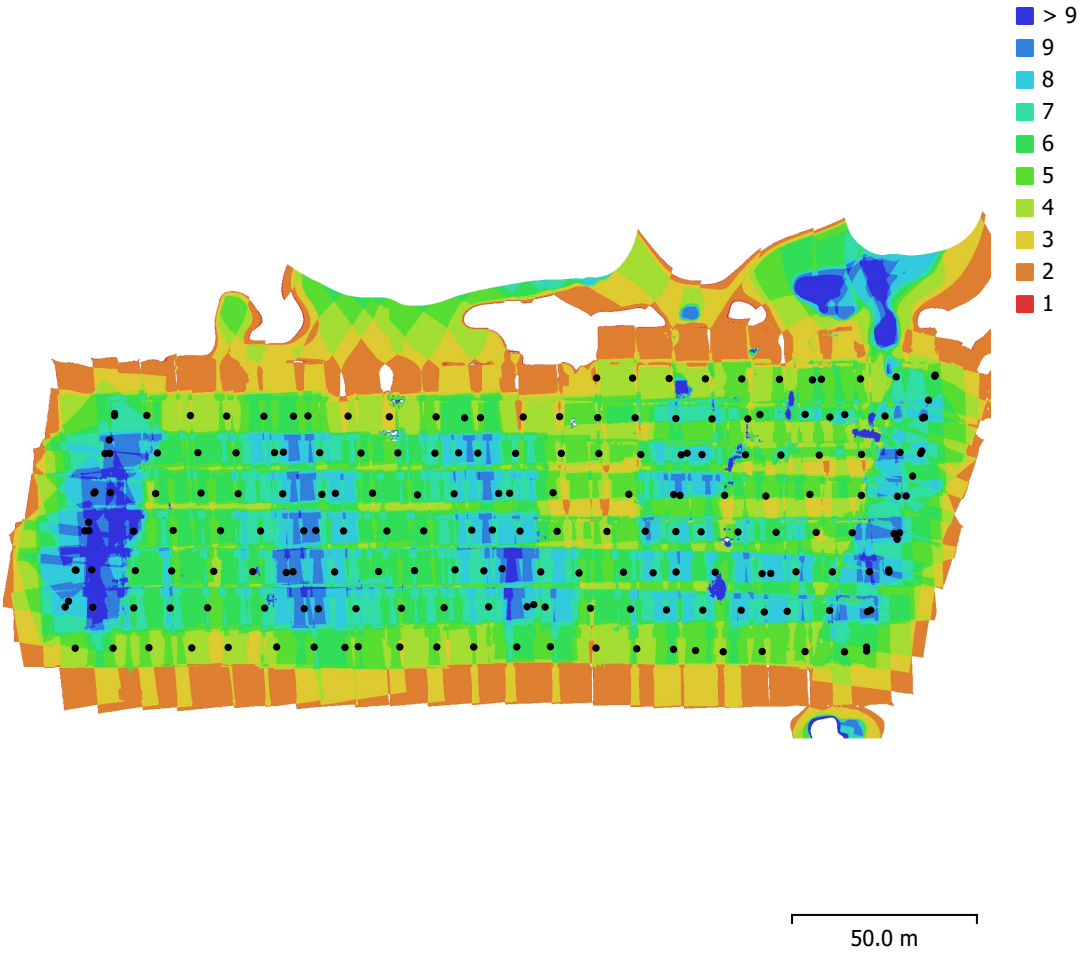


Fig. 1. Camera locations and image overlap.

Number of images:	200	Camera stations:	198
Flying altitude:	22.5 m	Tie points:	92,141
Ground resolution:	3.60 mm/pix	Projections:	213,136
Coverage area:	0.0276 km²	Reprojection error:	0.947 pix

Camera Model	Resolution	Focal Length	Pixel Size	Precalibrated
MAVIC2-ENTERPRISE-A...	8000 x 6000	4.5 mm	0.811 x 0.811 µm	No

Table 1. Cameras.

Camera Calibration

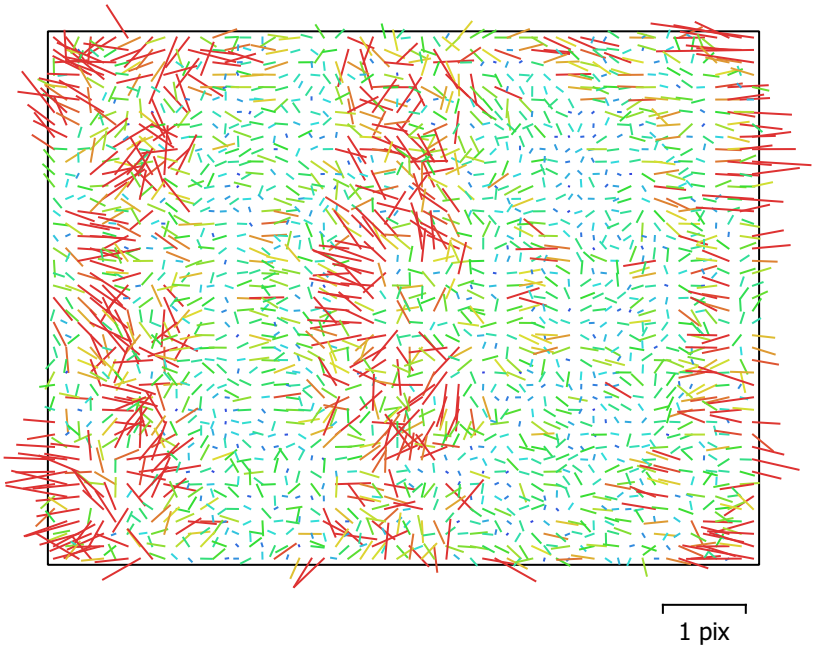


Fig. 2. Image residuals for MAVIC2-ENTERPRISE-ADVANCED (4.5mm).

MAVIC2-ENTERPRISE-ADVANCED (4.5mm)

200 images

Type	Resolution	Focal Length	Pixel Size
Frame	**8000 x 6000**	**4.5 mm**	**0.811 x 0.811 µm**

	Value	Error	F	Cx	Cy	B1	B2	K1	K2	P1	P2
F	**5692.65**	9.9	1.00	-0.31	0.70	-0.04	0.03	0.04	-0.84	0.05	-0.06
Cx	**4.66683**	0.53		1.00	-0.03	0.13	-0.12	0.12	0.20	0.16	0.08
Cy	**-5.12542**	0.54			1.00	0.07	0.05	0.03	-0.59	0.02	0.16
B1	**-3.2808**	0.044				1.00	0.04	-0.02	0.04	-0.02	0.02
B2	**-1.16421**	0.041					1.00	-0.03	-0.02	0.01	-0.03
K1	**-0.0066044**	0.0002						1.00	-0.28	0.12	0.35
K2	**-0.0125011**	0.0001							1.00	-0.10	0.00
P1	**-5.11407e-05**	7.6e-06								1.00	0.03
P2	**-9.70264e-05**	9.5e-06									1.00

Table 2. Calibration coefficients and correlation matrix.

Camera Locations

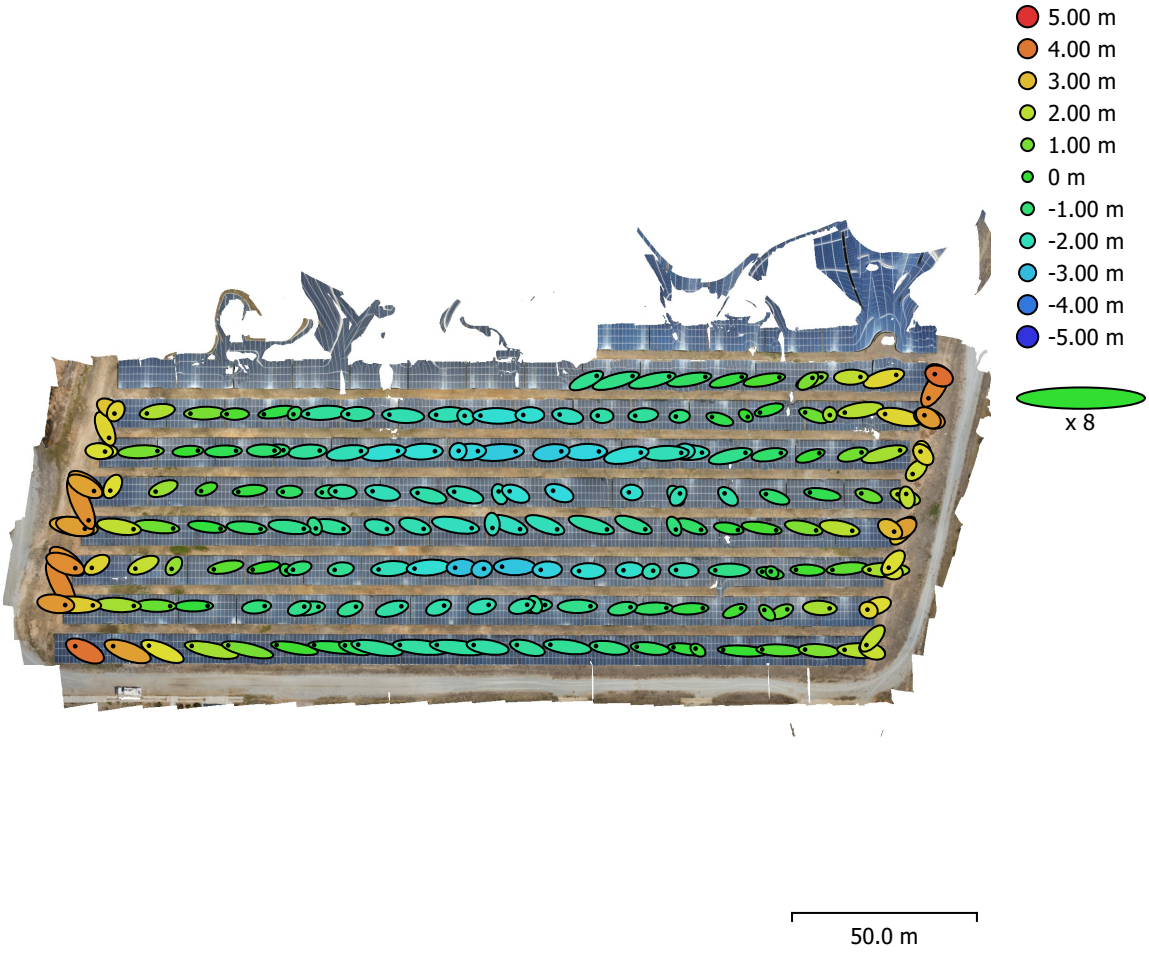


Fig. 3. Camera locations and error estimates.
Z error is represented by ellipse color. X,Y errors are represented by ellipse shape.
Estimated camera locations are marked with a black dot.

X error (m)	Y error (m)	Z error (m)	XY error (m)	Total error (m)
0.731107	0.208484	1.94311	0.760252	2.08654

Table 3. Average camera location error.
X - Longitude, Y - Latitude, Z - Altitude.

Digital Elevation Model

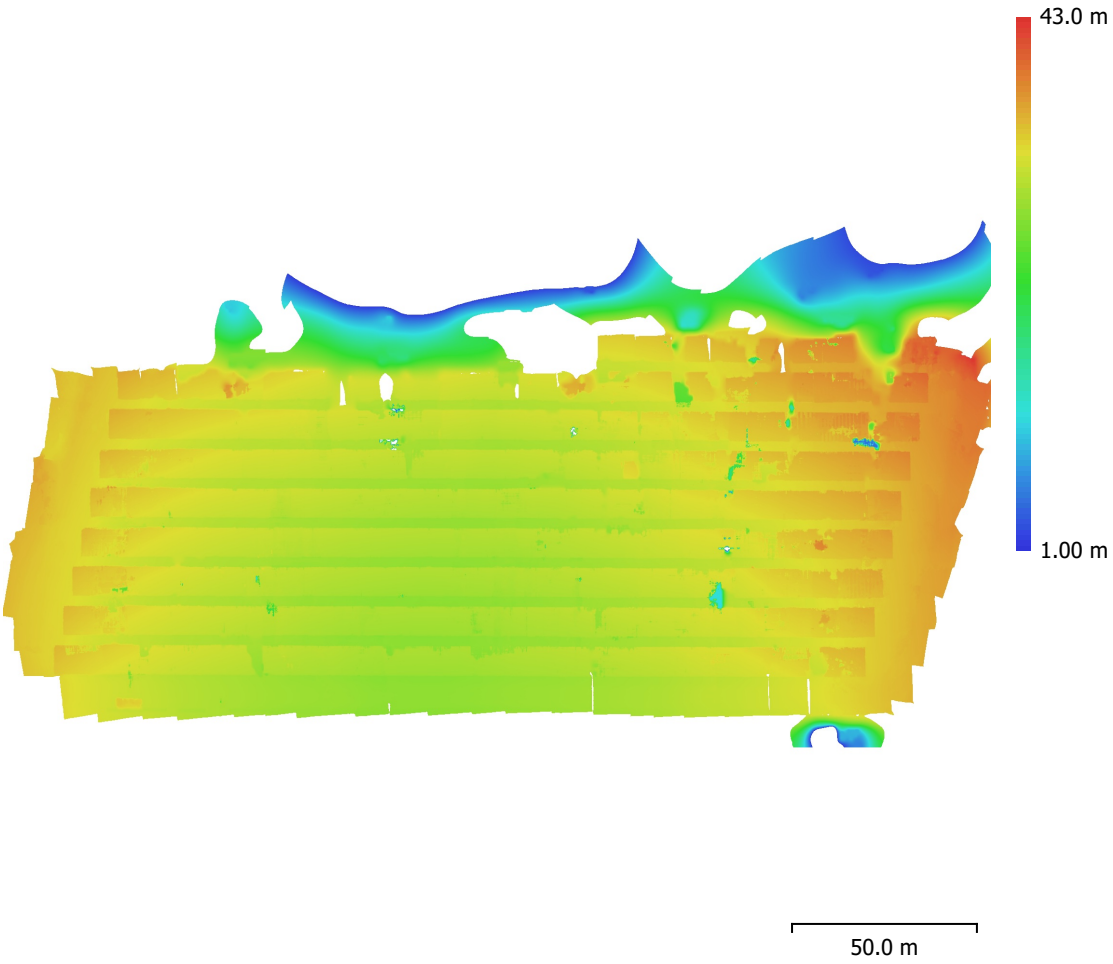


Fig. 4. Reconstructed digital elevation model.

Resolution:	7.89 mm/pix
Point density:	1.61 points/cm²

Processing Parameters

General	
Images	200
Aligned images	198
Coordinate system	WGS 84 (EPSG::4326)
Rotation angles	Yaw, Pitch, Roll
Tie Points	
Points	92,141 of 245,000
RMS reprojection error	0.294563 (0.946559 pix)
Max reprojection error	3.11818 (10.8336 pix)
Mean key point size	3.1269 pix
Point colors	3 bands, uint8
Key points	No
Average tie point multiplicity	2.52164
Alignment parameters	
Accuracy	High
Generic preselection	Yes
Reference preselection	Source
Key point limit	40,000
Key point limit per Mpx	1,000
Tie point limit	4,000
Exclude stationary tie points	Yes
Guided image matching	No
Adaptive camera model fitting	Yes
Matching time	18 minutes 52 seconds
Matching memory usage	432.15 MB
Alignment time	3 minutes 50 seconds
Alignment memory usage	274.24 MB
Optimization parameters	
Parameters	f, cx, cy, k1, k2, p1, p2
Adaptive camera model fitting	Yes
Exclude corners	No
Optimization time	6 seconds
Date created	2026:08:30 06:41:08
Software version	2.3.1.22580
File size	12.26 MB
Depth Maps	
Count	197
Depth maps generation parameters	
Quality	High
Filtering mode	Aggressive
Max neighbors	16
Processing time	1 hours 15 minutes
Memory usage	4.81 GB
Date created	2026:08:30 08:03:33
Software version	2.3.1.22580
File size	2.02 GB
Point Cloud	
Points	534,233,237
Coordinate precision	1.80 mm
Point attributes	

Color	3 bands, uint8
Normal	
Point classes	
Created (never classified)	534,233,237
Depth maps generation parameters	
Quality	High
Filtering mode	Aggressive
Max neighbors	16
Processing time	1 hours 15 minutes
Memory usage	4.81 GB
Point cloud generation parameters	
Source data	Depth maps
Processing time	41 minutes 51 seconds
Memory usage	9.69 GB
Date created	2026:08:30 08:45:37
Software version	2.3.1.22580
File size	6.88 GB
DEM	
Size	33,826 x 18,059
Resolution	7.89 mm/pix
Coordinate system	WGS 84 (EPSG::4326)
Depth maps generation parameters	
Quality	High
Filtering mode	Aggressive
Max neighbors	16
Processing time	1 hours 15 minutes
Memory usage	4.81 GB
Reconstruction parameters	
Source data	Depth maps
Interpolation	Enabled
Processing time	5 hours 4 minutes
Memory usage	10.10 GB
Date created	2026:08:30 18:37:44
Software version	2.3.1.22580
File size	1.80 GB
Orthomosaic	
Size	67,652 x 36,110
Resolution	3.95 mm/pix
Coordinate system	WGS 84 (EPSG::4326)
Colors	3 bands, uint8
Orthophotos	2.99 GB
Reconstruction parameters	
Blending mode	Mosaic
Surface	DEM
Enable hole filling	Yes
Enable ghosting filter	No
Processing time	41 minutes 10 seconds
Memory usage	12.43 GB
Date created	2026:08:30 18:51:54
Software version	2.3.1.22580
File size	4.57 GB
System	
Software name	Agisoft Metashape Professional
Software version	2.3.1 build 22580
OS	Windows 64 bit
RAM	31.72 GB

CPU	Intel(R) Core(TM) i7-8850H CPU @ 2.60GHz
GPU(s)	Quadro P1000